

1. Consider an MDP having three states S_1 , S_2 , and S_3 , with rewards -1 , -2 , and 0 for executing an action from that state, respectively. S_3 is the terminal state where only exit action is available that gives no reward. There is no other living reward. S_1 and S_2 each have two possible actions: a and b . 12
(CO1)
(PO1)

The transition function is described as follows:

- In S_1 , action a moves the agent to S_2 with probability 0.8 and makes the agent stay put with probability 0.2 .
- In S_2 , action a moves the agent to S_1 with probability 0.8 and makes the agent stay put with probability 0.2 .
- In either S_1 or S_2 , action b moves the agent to S_3 with probability 0.1 and makes the agent stay put with probability 0.9 .

Assume that the discount factor, $\gamma = 1$ and the initial policy has action b in both states.

With detailed steps, recommend the optimal policy for each state via policy iteration.

Solution:

Initially, $\pi(S) = b$ for $S \in \{S_1, S_2\}$. $R(S_1) = -1$, $R(S_2) = -2$, and $R(S_3) = 0$.

Policy Evaluation:

$$V(S_1) = -1 + 0.1 \times V(S_3) + 0.9 \times V(S_1)$$

$$V(S_2) = -2 + 0.1 \times V(S_3) + 0.9 \times V(S_2)$$

$$V(S_3) = 0$$

We get: $V(S_1) = -10$ and $V(S_2) = -20$.

Policy Improvement:

$$\begin{aligned} Q(S_1, a) &= 0.8 \times V(S_2) + 0.2 \times V(S_1) \\ &= -18 \end{aligned}$$

$$\begin{aligned} Q(S_1, b) &= 0.1 \times V(S_3) + 0.9 \times V(S_1) \\ &= -9 \end{aligned}$$

$$\begin{aligned} Q(S_2, a) &= 0.8 \times V(S_1) + 0.2 \times V(S_2) \\ &= -12 \end{aligned}$$

$$\begin{aligned} Q(S_2, b) &= 0.1 \times V(S_3) + 0.9 \times V(S_2) \\ &= -18 \end{aligned}$$

So the optimal policy:

$$\begin{aligned} \pi(S_1) &= \operatorname{argmax} (Q(S_1, a), Q(S_1, b)) \\ &= b \end{aligned}$$

$$\begin{aligned} \pi(S_2) &= \operatorname{argmax} (Q(S_2, a), Q(S_2, b)) \\ &= a \end{aligned}$$

Since the policy changed, we do another iteration:

Policy Evaluation:

$$V(S_1) = -1 + 0.1 \times V(S_3) + 0.9 \times V(S_1)$$

$$V(S_2) = -2 + 0.8 \times V(S_1) + 0.2 \times V(S_2)$$

$$V(S_3) = 0$$

We get: $V(S_1) = -10$ and $V(S_2) = -12.5$.

Policy Improvement:

$$\begin{aligned}Q(S_1, a) &= 0.8 \times V(S_2) + 0.2 \times V(S_1) \\&= -12\end{aligned}$$

$$\begin{aligned}Q(S_1, b) &= 0.1 \times V(S_3) + 0.9 \times V(S_1) \\&= -9\end{aligned}$$

$$\begin{aligned}Q(S_2, a) &= 0.8 \times V(S_1) + 0.2 \times V(S_2) \\&= -19.25\end{aligned}$$

$$\begin{aligned}Q(S_2, b) &= 0.1 \times V(S_3) + 0.9 \times V(S_2) \\&= -11.25\end{aligned}$$

So the optimal policy:

$$\begin{aligned}\pi(S_1) &= \operatorname{argmax} (Q(S_1, a), Q(S_1, b)) \\&= b\end{aligned}$$

$$\begin{aligned}\pi(S_2) &= \operatorname{argmax} (Q(S_2, a), Q(S_2, b)) \\&= a\end{aligned}$$

Since the policy remain unchanged, we can terminate. We get: $\pi(S_1) = b, \pi(S_2) = a, \pi(S_3) = EXIT$.

Rubric:

- 0.5 points for each value
- 0.5 points for each Q-value
- 0.5 points for each policy
- 1 point for each final policy

2. A meme is an image, video, piece of text, etc., typically humorous in nature, that is copied and spread rapidly by internet users, often with slight variations.

3

Create a meme related to this course. A text description will suffice.

Solution:

